

Introduction to Category Theory Through Algebra and Topology

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Chapter 1

Definitions and Examples

Abstract maybe, nonsense definitely.

Dmitry B. Fuchs

1.1 Introduction

The goal of this notes is to give an introduction to category theory via examples and motivations from abstract algebra and topology. We assume readers are familiar with first year graduate algebra (especially module theory, tensor product) and had at least one course in algebraic topology (that covers fundamental groups), although we will mainly work with groups, vector spaces, and topological spaces. We try to give as much examples as possible, and prove most of the statements we claimed. Hopefully, readers will find these notes easy to follow.

These notes organised as follows:

Chapter 1 covers the the definitions of category and functor.

Chapter 2 gives several important constructions in category theory, such as products, pull-backs, and limits.

Chapter 3 is still missing. Perhaps it will be about Abelian categories

Finally, these notes are always under construction.

1.2 Definition and Examples of Category

Definition 1.2.1 (Category). A **category** $\mathcal{C}$ consists of the following data:

CAT 1. a class $\text{Obj}(\mathcal{C})$ of objects. We write $X \in \text{Obj}(\mathcal{C})$ or $X \in \mathcal{C}$ to denote an object X in $\mathcal{C}$;

CAT 2. a class $\text{Mor}_{\mathcal{C}}(X, Y)$ of morphisms, for any objects X, Y in $\mathcal{C}$. We write $f : X \rightarrow Y$ to denote a morphism from X to Y , and sometimes write $\text{Mor}_{\mathcal{C}}(X, Y)$ instead of $\text{Mor}_{\mathcal{C}}(X, Y)$;

CAT 3. a composition law $\circ : \text{Mor}_{\mathcal{C}}(X, Y) \times \text{Mor}_{\mathcal{C}}(Y, Z) \rightarrow \text{Mor}_{\mathcal{C}}(X, Z)$.

These data must satisfy the following axioms:

Identity: for every object X , there exists an identity morphism $\text{id}_X \in \text{Mor}_{\mathcal{C}}(X, X)$ such that $f \circ \text{id}_X = f$ and $\text{id}_X \circ g = g$, for every morphisms $f : X \rightarrow Y$, and $g : Z \rightarrow X$;

Associativity: if $h : X \rightarrow Y$, $g : Y \rightarrow Z$, and $f : Z \rightarrow W$, then $(f \circ g) \circ h = f \circ (g \circ h)$.

Example 1.2.2. The following are categories:

Notation	Objects	Morphism	Composition
Set	sets	set functions	Usual way
Vect ($\mathbb{F}$)	$\mathbb{F}$ -vector spaces	$\mathbb{F}$ -linear maps	
Group	groups	group homomorphisms	
Top	topological spaces	continuous maps	

Example 1.2.3. Let G be a group, then it can be seen as a category $\underline{G}$ in the following way: objects of $\underline{G}$ consist of only the identity element e of G , and $\text{Mor}_{\underline{G}}(e, e) = G$. Composition is given by the group operation in G :

$$\begin{array}{ccc}
 e & \xrightarrow{g} & e \\
 & \searrow & \downarrow g' \\
 & & e
 \end{array}
 \quad .$$

$g' \circ g := g'g$

Example 1.2.4. Let X be a topological space. Define the **category $O(X)$ of opens in X** to have objects the open subsets $U \subseteq X$, and

$$\text{Mor}_{O(X)}(U, V) := \begin{cases} \{\text{inclusion } i : U \rightarrow V\} & \text{if } U \subseteq V, \\ \emptyset & \text{if else.} \end{cases}$$

Again, composition is performed in the usual way.

Example 1.2.5 (Opposite Category). Let $\mathcal{C}$ be a category. We can define a new category $\mathcal{C}^{op}$ by $\text{Obj}(\mathcal{C}^{op}) := \text{Obj}(\mathcal{C})$, and $\text{Mor}_{\mathcal{C}^{op}}(X, Y) := \text{Mor}_{\mathcal{C}}(Y, X)$.

In other words, morphisms in $\mathcal{C}^{op}$ are obtained by reversing the direction of morphisms in $\mathcal{C}$.

We call $\mathcal{C}^{op}$ the **opposite category** of $\mathcal{C}$.

Lemma 1.2.6 (Uniqueness of Identity Morphism). Let $\mathcal{C}$ be a category. For each $X, Y, Z \in \mathcal{C}$, there exists a unique morphism $\text{id}_X \in \text{Mor}_{\mathcal{C}}(X, X)$ such that $f \circ \text{id}_X = f$ and $\text{id}_X \circ g = g$, for every morphisms $f : X \rightarrow Y$, and $g : Z \rightarrow X$.

Proof. Suppose $\text{id}'_X \in \text{Mor}_{\mathcal{C}}(X, X)$ is another morphism satisfying $f' \circ \text{id}'_X = f'$ and $\text{id}'_X \circ g' = g'$, for every morphisms $f' : X \rightarrow Y$, and $g' : Z \rightarrow X$. Then we have

$$\begin{aligned} \text{id}_X &= \text{id}_X \circ \text{id}'_X && (\text{Put } f' = \text{id}_X) \\ &= \text{id}_X \circ \text{id}'_X && (\text{Put } g = \text{id}'_X) \\ &= \text{id}'_X. \end{aligned}$$

□

Definition 1.2.7 (Subcategory). A **subcategory** of a category $\mathcal{C}$ is a category $\mathcal{S}$ whose objects are objects in $\mathcal{C}$, and whose morphisms are morphisms in $\mathcal{C}$ with the same identity morphisms and composition law.

Example 1.2.8. **Group** is a subcategory of **Set**.

Example 1.2.9. The category $\mathbf{fgVect}(\mathbb{F})$ of finite dimensional $\mathbb{F}$ -vector spaces is a subcategory of $\mathbf{Vect}(\mathbb{F})$.

Example 1.2.10. The category **CW** of CW-complexes is a subcategory of **Top**.

Definition 1.2.11 (Small Category). A category $\mathcal{C}$ is **small** if the collection of objects and morphisms are sets.

Example 1.2.12. By the Russell's paradox, we know **Set** is not a small category.

Example 1.2.13. For any group G , the associated category $\underline{G}$ is a small category.

Small categories are extremely important in simplicial homotopy theory.

Definition 1.2.14 (Isomorphism). Let $\mathcal{C}$ be a category. We say $f \in \text{Mor}_{\mathcal{C}}(X, Y)$ is an **isomorphism** if there exists an **inverse (morphism)** $f^{-1} \in \text{Mor}_{\mathcal{C}}(Y, X)$ such that $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$.

Lemma 1.2.15. Inverse morphism, if exists, is unique.

Proof. Exercise. □

Example 1.2.16. In Example 1.2.2, the categorical isomorphisms are the isomorphisms in the usual sense.

Example 1.2.17. Let **hTop** be the homotopy category of topological spaces, whose objects are the same as **Top**, and morphisms are homotopy classes of maps. Then an isomorphism in **hTop** is a homotopy equivalence.

1.3 Functors

Definition 1.3.1 (Covariant and Contravariant Functor). A covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is a mapping between categories such that

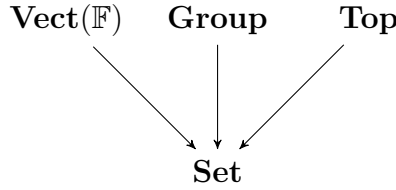
- (1) $F(X) \in \mathcal{D}$ for every $X \in \mathcal{C}$.
- (2) $F(\varphi) \in \text{Mor}_{\mathcal{D}}(F(X), F(Y))$ for every $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$.
- (3) $F(\text{id}_X) = \text{id}_{F(X)}$ for every $X \in \mathcal{C}$.
- (4) If $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$ and $\psi \in \text{Mor}_{\mathcal{C}}(Y, Z)$, then $F(\psi \circ \varphi) = F(\psi) \circ F(\varphi)$.

A **contra-variant functor** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a covariant functor $F : \mathcal{C}^{op} \rightarrow \mathcal{D}$. In particular, we have $F(\varphi) \in \text{Mor}_{\mathcal{D}}(F(Y), F(X))$ for every $\varphi \in \text{Mor}_{\mathcal{C}}(X, Y)$, and $F(\psi \circ \varphi) = F(\varphi) \circ F(\psi)$.

Throughout this notes, the word "functor" will always mean a covariant functor.

There are lots of examples of functors from algebra and topology.

Example 1.3.2. We have **forgetful functors**



defined by dropping the structures of objects in the source category.

Example 1.3.3. Recall the definition of the torsion subgroup of a group G

$$\text{Tors}(G) := \{g \in G \mid g^n = e \text{ for some } n \geq 0\}.$$

Then it defines a functor $\text{Tors}(-) : \mathbf{Group} \rightarrow \mathbf{Group}$.

Example 1.3.4. Fix $M \in \mathbf{Vect}(\mathbb{F})$, then for any $N \in \mathbf{Vect}(\mathbb{F})$ one can consider mappings

$$\begin{aligned} N &\mapsto N \otimes_{\mathbb{F}} M, \\ N &\mapsto \text{Hom}_{\mathbf{Vect}(\mathbb{F})}(N, M). \end{aligned}$$

The first one defines a functor $- \otimes_{\mathbb{F}} M : \mathbf{Vect}(\mathbb{F}) \rightarrow \mathbf{Vect}(\mathbb{F})$, while the second one $\text{Hom}_{\mathbf{Vect}(\mathbb{F})}(-, M)$ is contra-variant.

Example 1.3.5. Topologists know the fundamental group and classifying spaces define functors

$$\begin{aligned}\pi_1(-) : \mathbf{Top} &\rightarrow \mathbf{Group}, \\ B- : \mathbf{Group} &\rightarrow \mathbf{Top}.\end{aligned}$$

Note that these functors are not inverses, since $\pi_1(\mathbb{R}) = \pi_1(\mathbb{CP}^\infty) = 0$, and $\mathbb{R}$ and $\mathbb{CP}^\infty$ are not homotopy equivalent, since $\pi_2(\mathbb{CP}^\infty) \cong \mathbb{Z}$.

Example 1.3.6. A **presheaf** of abelian groups on a topological space X is a functor $\mathcal{F} : O(X)^{op} \rightarrow \mathbf{Ab}$ from the opposite category of $O(X)$ to the category of abelian groups.

Exercise 1.3.7. Show that the category **CAT** of categories and functors is indeed a category.

1.3.1 Natural Transformation

Functors provide bridges between categories, natural transformations provide bridges between functors.

Definition 1.3.8. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. A **natural transformation** $\eta : F \rightarrow G$ from F to G is a collection $\{\eta_X \mid X \in \mathcal{C}\}$ of morphisms in $\mathcal{D}$ such that the diagram

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \eta_X \downarrow & & \downarrow \eta_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

commutes, for every $f \in \text{Mor}_{\mathcal{C}}(X, Y)$.

Example 1.3.9. Let **Ring** be the category of commutative unital rings. We have two functors

$$\begin{aligned} \mathbf{Ab} &\xleftarrow{\quad \times \quad} \mathbf{Ring} \xrightarrow{GL(n, -)} \mathbf{Ab} \\ R^\times &\longleftarrow R \longrightarrow GL(n, R) \end{aligned}$$

for which $R^\times$ is the multiplicative group of units in R . The determinant is a natural transformation from $GL(n, -)$ to $\times$.

Example 1.3.10. Fix two non-negative integers n, q . Let π, G be two Abelian groups. Then we have two functors

$$H^n(-; \pi), H^q(-; G) : \mathbf{Top} \rightarrow \mathbf{Ab}$$

mapping a topological space to its cohomology groups. A **cohomology operation** of type (n, q, π, G) is a natural transformation $\varphi : H^n(-; \pi) \rightarrow H^q(-; G)$.

Denote by $\mathcal{O}(n, q, \pi, G)$ the set of cohomology operations of type (n, q, π, G) . Since G is an Abelian group, $\mathcal{O}(n, q, \pi, G)$ is an Abelian group (under point-wise addition). In particular, there is an isomorphism

$$\mathcal{O}(n, q, \pi, G) \cong H^q(K(\pi, n); G),$$

for which $K(\pi, n)$ is the Eilenberg-MacLane space.

Exercise 1.3.11. Let $\mathcal{C}, \mathcal{D}$ be categories. Prove that the category of functors from $\mathcal{C} \rightarrow \mathcal{D}$, and natural transformations is indeed a category.

1.3.2 Adjoint Functors

Definition 1.3.12 (Adjoint Functors). we say two functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$ are **adjoint** if there exists a **natural bijection** between morphisms

$$\mathrm{Hom}_{\mathcal{C}}(G(D), C) \cong \mathrm{Hom}_{\mathcal{D}}(D, F(C)),$$

for every $C \in \mathcal{C}$ and $D \in \mathcal{D}$.

In this case, we say G is **left adjoint** to F , and F is **right adjoint** to G .

The **naturality** of the bijection

$$\varphi : \mathrm{Hom}_{\mathcal{C}}(G(D), C) \rightarrow \mathrm{Hom}_{\mathcal{D}}(D, F(C))$$

means the following two conditions

(i). Suppose $f : D \rightarrow D'$ is a morphism in $\mathcal{D}$, then the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{C}}(G(D'), C) & \xrightarrow{(Gf)^*} & \mathrm{Hom}_{\mathcal{C}}(G(D), C) \\ \varphi \downarrow & & \downarrow \varphi \\ \mathrm{Hom}_{\mathcal{D}}(D', F(C)) & \xrightarrow{f^*} & \mathrm{Hom}_{\mathcal{D}}(D, F(C)) \end{array}$$

commutes; and

(ii). if $g : C \rightarrow C'$ is a morphism in $\mathcal{C}$, then the diagram

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{C}}(G(D), C) & \xrightarrow{g_*} & \mathrm{Hom}_{\mathcal{C}}(G(D), C') \\ \varphi \downarrow & & \downarrow \varphi \\ \mathrm{Hom}_{\mathcal{D}}(D, F(C)) & \xrightarrow{(Fg)_*} & \mathrm{Hom}_{\mathcal{D}}(D, F(C')) \end{array}$$

commutes.

Loosely speaking, a bijection is natural means its definition is independent of the choice of objects.

Example 1.3.13 (Fundamental Theorem of Tensor Product). Let R, S be rings, and fix an (R, S) -bimodule X . Then the functors $- \otimes_R X, \text{Hom}_S(X, -)$ are adjoint, that is, there exists a natural bijection

$$\text{Hom}_S(Y \otimes_R X, Z) \cong \text{Hom}_R(Y, \text{Hom}_S(X, Z))$$

for every R -module Y , and S -module Z . To see this, we define $\Phi : \text{Hom}_S(Y \otimes_R X, Z) \rightarrow \text{Hom}_R(Y, \text{Hom}_S(X, Z))$ by

$$[\Phi(f)(y)](x) := f(y \otimes x),$$

and $\Psi : \text{Hom}_R(Y, \text{Hom}_S(X, Z)) \rightarrow \text{Hom}_S(Y \otimes_R X, Z)$ by

$$[\Psi(g)](y \otimes x) := [g(y)](x).$$

Readers can verify that they are inverses and are natural.

Example 1.3.14. Given $S \in \mathbf{Set}$, one can form the free Abelian group $\mathbb{Z}\langle S \rangle$ generated by S . This defines a functor

$$\mathbb{Z}\langle - \rangle : \mathbf{Set} \rightarrow \mathbf{Ab},$$

which is right adjoint to the forgetful functor $U : \mathbf{Ab} \rightarrow \mathbf{Set}$.

Example 1.3.15. Let $X \in \mathbf{Top}$, then one can form the suspension ΣX over X :

$$\Sigma X := X \times I / \sim,$$

for which $(x, 1) \sim (y, 1)$, and $(x, 0) \sim (y, 0)$ for all $x, y \in X$. One can also form the loop space

$$\begin{aligned} \Omega X &:= \text{Map}_*(\mathbb{S}^1, X) \\ &:= \{ \text{continuous } f : \mathbb{S}^1 \rightarrow X \mid f \text{ preserves base-point} \} \end{aligned}$$

Then the functors Σ and Ω are adjoint:

$$\text{Map}_*(\Sigma X, Y) \cong \text{Map}_*(X, \Omega Y).$$

Indeed, we define $\Phi : \text{Map}_*(\Sigma X, Y) \rightarrow \text{Map}_*(X, \Omega Y)$ by

$$[\Phi(f)(x)](t) := f[x, t];$$

and define $\Psi : \text{Map}_*(X, \Omega Y) \rightarrow \text{Map}_*(\Sigma X, Y)$ by

$$\Psi(g)[x, t] := (g(x))(t).$$

We leave it to the readers to check Φ, Ψ are inverses and are natural.

Chapter 2

Constructions in Categories

Its easy.

Vitaly O. Tarasov

2.1 Product and Coproduct

2.1.1 Product

Definition 2.1.1 (Product). Let $\mathcal{C}$ be a category. The **product** of two objects A, B in $\mathcal{C}$ is an object $A \times B$ in $\mathcal{C}$, together with two morphisms

$$\begin{aligned}\pi_1 : A \times B &\rightarrow A, \\ \pi_2 : A \times B &\rightarrow B.\end{aligned}$$

that satisfying the following universal property: suppose X is another object in $\mathcal{C}$ together with two morphisms $f_1 : X \rightarrow A$, $f_2 : X \rightarrow B$, then there exists a unique morphism $f : X \rightarrow A \times B$ such that the diagram

$$\begin{array}{ccccc} & & X & & \\ & f_1 \swarrow & & \searrow f_2 & \\ A & & A \times B & & B \\ & \xleftarrow{\pi_1} & & \xrightarrow{\pi_2} & \end{array}$$

$\downarrow f$

commutes.

Example 2.1.2. The product of two sets A, B is precisely their Cartesian product $A \times B$, hence justifying our notation. The maps π_1, π_2 are just canonical projections.

We verify the universal property here. Suppose X is another set, together with two maps $f_1 : X \rightarrow A$, $f_2 : X \rightarrow B$. Define $f : X \rightarrow A \times B$ by

$$f(x) := (f_1(x), f_2(x)).$$

Then clearly the diagram

$$\begin{array}{ccccc} & & X & & \\ & f_1 \swarrow & \downarrow f & \searrow f_2 & \\ A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B \end{array}$$

commutes. It remains to the uniqueness of f .

Suppose $g : X \rightarrow A \times B$ is another map that makes the above diagram commutes. We then know g is given by

$$g(x) = (g_1(x), g_2(x)),$$

for some functions $g_1 : X \rightarrow A$, $g_2 : X \rightarrow B$. Using the commutativity of the above diagram, we immediately see that $f_i = g_i$ for $i = 1, 2$. Thus $f = g$.

Example 2.1.3. A slight modification on Example 2.1.2 shows that the product of two topological spaces X, Y is the product space $X \times Y$ equipped with the product topology.

Example 2.1.4. Product may not exist in a category. Consider the category **Fields** of fields. One can check directly that the Cartesian product $F_1 \times F_2$ of two fields F_1 and F_2 together with the canonical projections satisfy the universal property. However, $F_1 \times F_2$ is not a field – it is not an integral domain.

Proposition 2.1.5. If the product exists in the category, then it is unique up to natural isomorphism.

Proof. Fix $A, B \in \mathcal{C}$ such that their product $A \times B$ exists. Suppose X is another object in $\mathcal{C}$ that satisfies the universal property of product. Then combining the Universal Properties of X and $A \times B$, we get a commutative diagram

$$\begin{array}{ccccc} & & X & & \\ & f_1 \swarrow & \downarrow g & \searrow f_2 & \\ A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B \end{array}$$

In particular, this gives the following commutative diagram:

$$\begin{array}{ccccc}
 & & A \times B & & \\
 & \swarrow \pi_1 & \downarrow fg & \searrow \pi_2 & \\
 A & \xleftarrow{\pi_1} & A \times B & \xrightarrow{\pi_2} & B.
 \end{array}$$

Now, if we replace fg by $\text{id}_{A \times B}$, the diagram still commute. Thus by the uniqueness part of the universal property of $A \times B$, we must have $fg = \text{id}_{A \times B}$.

Similarly, we have another commutative diagram:

$$\begin{array}{ccccc}
 & & X & & \\
 & \swarrow f_1 & \downarrow gf & \searrow f_2 & \\
 A & \xleftarrow{f_1} & X & \xrightarrow{f_2} & B.
 \end{array}$$

The exact same argument shows $gf = \text{id}_X$. As a result, $X \cong A \times B$. □

Corollary 2.1.6. Product is commutative up to isomorphism.

Proposition 2.1.7. Product is associative up to natural isomorphism:

$$(A \times B) \times C \cong A \times (B \times C).$$

Proof. Consider the following diagram

$$\begin{array}{ccccc}
 A & \xleftarrow{p_1} & A \times B & \xrightarrow{p_2} & B \\
 \uparrow s_1 & & \uparrow & & \uparrow q_1 \\
 A \times (B \times C) & \xrightarrow{s_2} & B \times C & & \\
 \downarrow r_1 & & \downarrow q_2 & & \\
 (A \times B) \times C & \xrightarrow{r_2} & C & &
 \end{array}$$

We have $s_1 : A \times (B \times C) \rightarrow A$, and $q_1 s_2 : A \times (B \times C) \rightarrow B$. So by the universal property of $A \times B$, there exists a unique map $h : A \times (B \times C) \rightarrow A \times B$ such that the diagram

$$\begin{array}{ccccc}
A & \xleftarrow{p_1} & A \times B & \xrightarrow{p_2} & B \\
s_1 \swarrow & & \uparrow h & & \nwarrow q_1 \\
A \times (B \times C) & \xrightarrow{s_2} & B \times C & & \\
\downarrow r_1 & & & & \downarrow q_2 \\
(A \times B) \times C & \xrightarrow{r_2} & C & &
\end{array}$$

commutes. Then we have $h : A \times (B \times C) \rightarrow A \times B$, and $q_2 s_2 : A \times (B \times C) \rightarrow C$. So by the universal property of $(A \times B) \times C$, we have a map $i : A \times (B \times C) \rightarrow (A \times B) \times C$ such that the diagram

$$\begin{array}{ccccc}
A & \xleftarrow{p_1} & A \times B & \xrightarrow{p_2} & B \\
s_1 \swarrow & & \uparrow h & & \nwarrow q_1 \\
A \times (B \times C) & \xrightarrow{s_2} & B \times C & & \\
\downarrow r_1 & & & & \downarrow q_2 \\
(A \times B) \times C & \xrightarrow{r_2} & C & &
\end{array}$$

commutes. Using the same diagram chasing method, one can show there exists maps $g : (A \times B) \times C \rightarrow A \times (B \times C)$, $f : (A \times B) \times C \rightarrow B \times C$, such that the diagram

$$\begin{array}{ccccc}
A & \xleftarrow{p_1} & A \times B & \xrightarrow{p_2} & B \\
s_1 \swarrow & & \uparrow h & & \nwarrow q_1 \\
A \times (B \times C) & \xrightarrow{s_2} & B \times C & & \\
\downarrow r_1 & & & & \downarrow q_2 \\
(A \times B) \times C & \xrightarrow{r_2} & C & &
\end{array}$$

commutes. We wish to show $gi = \text{id}_{(A \times B) \times C}$.

Firstly, we have

$$\begin{aligned}
(q_1 s_2)gi &= q_1 f i \\
&= q_1 s_2,
\end{aligned}$$

and

$$\begin{aligned}(q_2 s_2)gi &= q_2 fi \\ &= q_2 s_2.\end{aligned}$$

Thus using the same logic in the proof of Proposition 2.1.5, we have $gi = \text{id}_{(A \times B) \times C}$, and similarly $ig = \text{id}_{A \times (B \times C)}$. Hence $A \times (B \times C) \cong (A \times B) \times C$. $\square$

2.1.2 Coproduct

Definition 2.1.8 (Coproduct). Let $\mathcal{C}$ be a category. The **coproduct** of two objects A, B in $\mathcal{C}$ is another object $A \sqcup B$ in $\mathcal{C}$, together with two morphisms

$$\begin{aligned}i_1 : A &\rightarrow A \sqcup B, \\ i_2 : B &\rightarrow A \sqcup B.\end{aligned}$$

that satisfying the following universal property: suppose X is another object in $\mathcal{C}$ together with two morphisms $f_1 : A \rightarrow X$, $f_2 : B \rightarrow X$, then there exists a unique morphism $f : A \sqcup B \rightarrow X$ such that the diagram

$$\begin{array}{ccccc} & & X & & \\ & \nearrow f_1 & \uparrow f & \nwarrow f_2 & \\ A & \xrightarrow{i_1} & A \sqcup B & \xleftarrow{i_2} & B \end{array}$$

commutes.

Perhaps it is worth to point out that the definition of coproduct is just reversing all arrows in the definition of product.

Proposition 2.1.9. If the coproduct exists in the category, then it is unique up to natural isomorphism.

Proof. Exercise, c.f. the proof of Proposition 2.1.5. $\square$

Corollary 2.1.10. Coproduct is commutative up to isomorphism.

Proposition 2.1.11. Coproduct is associative up to natural isomorphism:

$$(A \sqcup B) \sqcup C \cong A \sqcup (B \sqcup C).$$

Proof. Exercise, c.f. the proof of Proposition 2.1.7. $\square$

Example 2.1.12. The coproduct of two sets A, B is their disjoint union.

Example 2.1.13. The coproduct of two groups A, B is their free product $A * B$. The coproduct of two Abelian groups C, D , however, is their direct sum $C \oplus D$. This is because the free product of two Abelian groups is not necessarily free. For instance, consider $C = D = \mathbb{Z}$, the free group of one generator, which is also Abelian. Then their free product F_2 is the free group of two generators. Topologists know that F_2 is not Abelian – it is the fundamental group of the figure eight.

2.1.3 Product and Coproduct in the Category of Vector Spaces

Let V, W be finite-dimensional $\mathbb{F}$ -vector spaces, then one can show that $V \times W \cong V \oplus W$. The point of insisting finite-dimensional is to make sure V and W are not a product (or coproduct) of infinitely many factors. The goal of this section is to study this situation carefully – we would like to understand the vector space structure of a product and coproduct of infinitely many vector spaces.

The case for product is natural:

Proposition 2.1.14. Let I be an infinite index set, and $V_i \in \mathbf{Vect}(\mathbb{F})$ for each $i \in I$. Then $\prod_{i \in I} V_i$, as a set, is given by

$$\prod_{i \in I} V_i = \{(v_i)_{i \in I} \mid v_i \in V_i\}.$$

The vector addition and scalar multiplication are performed coordinate-wise.

Proof. This is just a straight verification of the universal property of product. $\square$

We would like to point out that this is not the coproduct. Take $I = \mathbb{N}$, $V_i := \mathbb{F}$ for each i , and let $V := \prod_{i \in I} V_i$. We can map V_i into V via the canonical inclusion

$$\varphi_i : x \mapsto \left(0, \dots, 0, \underbrace{x}_i, 0, \dots \right).$$

This allows us to consider the following coproduct diagram:

$$\begin{array}{ccccc} & & \mathbb{F} & & \\ & \nearrow \text{id}_{\mathbb{F}} & \uparrow f & \nwarrow \text{id}_{\mathbb{F}} & \\ V_i & \xrightarrow{\varphi_i} & V & \xleftarrow{\varphi_j} & V_j \end{array}$$

We remind readers that if V were the coproduct, then there exists a unique map $f : V \rightarrow \mathbb{F}$, such that $f\varphi_i = \text{id}_{\mathbb{F}}$ for all i . If $i = 1$, then f must be the projection π_1 onto the first factor; if $i = 2$, then $f = \pi_2$. This is absurd as $\pi_1 \neq \pi_2$. So V is not the coproduct.

We now give a description of coproduct of infinitely many vector spaces:

Proposition 2.1.15. Let I be an infinite index set, and $V_i \in \mathbf{Vect}(\mathbb{F})$ for each $i \in I$.

Then $\bigoplus_{i \in I} V_i$, as a set, is given by

$$\bigoplus_{i \in I} V_i = \left\{ (v_i)_{i \in I} \in \prod_{i \in I} V_i \mid v_i = 0 \text{ except for finitely many } i \in I \right\}.$$

The vector addition and scalar multiplication are performed coordinate-wise.

Proof. Let $W := \bigoplus_{i \in I} V_i$ as described. Again, we have the canonical inclusion $\varphi_i : V_i \rightarrow W$ given by

$$\varphi_i(x) := (v_j)_{j \in I},$$

for which

$$v_j = \begin{cases} x & \text{if } j = i, \\ 0 & \text{if else.} \end{cases}$$

If W' is another $\mathbb{F}$ -vector space with maps $f_i : V_i \rightarrow W'$ for all $i \in I$, then we define an $\mathbb{F}$ -linear map $f : W \rightarrow W'$ by

$$f((v_i)_{i \in I}) := \sum_{i \in I} f_i(v_i).$$

Since we assume there can only be finitely many non-zero v_i 's, there is no convergent issue, and we have the commutative diagram

$$\begin{array}{ccccc} & & W' & & \\ & f_i \nearrow & \uparrow f & \nwarrow f_j & \\ V_i & \xrightarrow{\varphi_i} & W & \xleftarrow{\varphi_j} & V_j \end{array}$$

for all $i, j \in I$.

It remains to show f is unique. If $f' : W \rightarrow W'$ is another $\mathbb{F}$ -linear map such that the diagram

$$\begin{array}{ccccc}
 & & W' & & \\
 & f_i \nearrow & \uparrow f' & \nwarrow f_j & \\
 V_i & \xrightarrow{\varphi_i} & W & \xleftarrow{\varphi_j} & V_j
 \end{array}$$

commutes, then let e_i be the i -th canonical basis vector of W . We see that

$$\begin{aligned}
 f'(e_i) &= f'\varphi_i(1) \\
 &= f_i(1) \\
 &= f\varphi_i(1) \\
 &= f(e_i).
 \end{aligned}$$

Therefore, $f = f'$ as desired. □

2.2 Pullback and Pushout

2.2.1 Pullback

Definition 2.2.1 (Pullback). The **pullback** of two morphisms $f : A \rightarrow C$, $g : B \rightarrow C$ in a category $\mathbf{C}$ is an object $A \times_C B$, together with two morphisms $\pi_1 : A \times_C B \rightarrow A$, $\pi_2 : A \times_C B \rightarrow B$, such that the diagram

$$\begin{array}{ccc}
 A \times_C B & \xrightarrow{\pi_2} & B \\
 \pi_1 \downarrow & & \downarrow g \\
 A & \xrightarrow{f} & C
 \end{array}$$

commutes. In addition, the data $(A \times_C B, \pi_1, \pi_2)$ satisfies the following universal property: if X is another object in $\mathbf{C}$ with two morphisms π'_1, π'_2 , such that the diagram

$$\begin{array}{ccc}
 X & \xrightarrow{\pi'_2} & B \\
 \pi'_1 \downarrow & & \downarrow g \\
 A & \xrightarrow{f} & C
 \end{array}$$

commutes, then there exists a unique morphism $s : X \rightarrow A \times_C B$ so that the diagram

$$\begin{array}{ccccc}
X & & \xrightarrow{\pi'_2} & & B \\
& \searrow s & & \nearrow \pi_2 & \\
\pi'_1 \downarrow & & A \times_C B & & \downarrow g \\
& \nearrow \pi_1 & & & \\
A & & \xrightarrow{f} & & C
\end{array}$$

commutes.

Example 2.2.2. The pullback of two functions $f : A \rightarrow C$, $g : B \rightarrow C$ in **Set** is the **fibre product**

$$A \times_C B := \{(a, b) \in A \times B \mid f(a) = g(b)\},$$

together with the canonical projections $\pi_1 : A \times_C B \rightarrow A$, $\pi_2 : A \times_C B \rightarrow B$.

Example 2.2.3. Let X, Y be topological spaces, and $p : E \rightarrow X$ be a vector bundle. If $f : Y \rightarrow X$ is a continuous map, then one can form the **induced bundle**

$$f^*E := \{(y, e) \in Y \times E \mid f(y) = p(e)\}.$$

The projection map $\pi : f^*E \rightarrow Y$ gives a bundle structure over Y . This is the pullback in the category **VectBun** of vector bundles.

Proposition 2.2.4. Given a commutative ladder

$$\begin{array}{ccccc}
A & \xrightarrow{f_1} & B & \xrightarrow{f_2} & C \\
\downarrow i & & \downarrow j & & \downarrow k \\
D & \xrightarrow{g_1} & E & \xrightarrow{g_2} & F
\end{array}$$

in a category **C**, for which each square is a pull-back. Then the outer rectangle is also a pull-back.

Proof. We only need to verify the universal property of Pullback.

Suppose X is an object in **C**, with two morphisms $p : X \rightarrow C$, $q : X \rightarrow D$ such that the diagram

$$\begin{array}{ccc}
 X & \xrightarrow{p} & C \\
 q \downarrow & & \downarrow k \\
 D & \xrightarrow{g_2 g_1} & F
 \end{array}$$

commutes. Then it gives another commutative diagram:

$$\begin{array}{ccc}
 X & \xrightarrow{p} & C \\
 g_1 q \downarrow & & \downarrow k \\
 E & \xrightarrow{g_2} & F
 \end{array}$$

Since B is a pullback, the universal property asserts that there exists a unique map $\alpha : X \rightarrow B$ such that the diagram

$$\begin{array}{ccccc}
 X & \xrightarrow{p} & C & & \\
 \searrow \alpha & & \nearrow f_2 & & \\
 & B & & & \\
 \nearrow j & & \searrow g & & \\
 E & \xrightarrow{f} & F & & \\
 \uparrow g_1 q & & & &
 \end{array}$$

commutes. This gives a new commutative diagram:

$$\begin{array}{ccc}
 X & \xrightarrow{\alpha} & B \\
 q \downarrow & & \downarrow j \\
 D & \xrightarrow{g_1} & E
 \end{array}$$

Finally, since A is a pullback, the universal property yields a morphism $\beta : X \rightarrow A$ so that the diagram

$$\begin{array}{ccc}
X & \xrightarrow{\alpha} & B \\
q \downarrow & \searrow \beta & \nearrow f_1 \\
& & A \\
& \nearrow i & \downarrow j \\
D & \xrightarrow{g_1} & E
\end{array}$$

commutes, and hence complete the proof. $\square$

Pullback is important for topologists, because of the Splitting Principle:

Theorem 2.2.5 (Splitting Principle). If $E \rightarrow B$ is a vector bundle, then there exists a map $f : B' \rightarrow B$ such that

- (1) f^*E is a Whitney sum of line bundles;
- (2) the induced map $f^* : H^*(B) \rightarrow H^*(B')$ between cohomology rings is injective.

2.2.2 Pushout

Definition 2.2.6 (Pushout). The **pushout** of two morphisms $f : C \rightarrow A$, $g : C \rightarrow B$ in a category $\mathbf{C}$ is an object $A \sqcup_C B$, together with two morphisms $i_1 : A \rightarrow A \sqcup_C B$, $i_2 : B \rightarrow A \sqcup_C B$, such that the diagram

$$\begin{array}{ccc}
C & \xrightarrow{f} & A \\
g \downarrow & & \downarrow i_1 \\
B & \xrightarrow{i_2} & A \sqcup_C B
\end{array}$$

commutes. In addition, the data $(A \sqcup_C B, i_1, i_2)$ satisfies the following universal property: if X is another object in $\mathbf{C}$ with two morphisms i'_1, i'_2 , such that the diagram

$$\begin{array}{ccc}
C & \xrightarrow{f} & A \\
g \downarrow & & \downarrow i'_1 \\
B & \xrightarrow{i'_2} & X
\end{array}$$

commutes, then there exists a unique morphism $s : A \sqcup_C B \rightarrow X$ so that the diagram

$$\begin{array}{ccc}
C & \xrightarrow{f} & A \\
g \downarrow & \nearrow i_1 & \downarrow i'_1 \\
& A \sqcup_C B & \\
& \nearrow i_2 \quad \searrow s & \\
B & \xrightarrow{i'_2} & X
\end{array}$$

commutes.

Example 2.2.7. The pushout of two functions $f : C \rightarrow A$, $g : C \rightarrow B$ is the set

$$A \sqcup_C B := (A \sqcup B) / \sim,$$

together with the canonical inclusions $i_1 : A \rightarrow A \sqcup_C B$, $i_2 : B \rightarrow A \sqcup_C B$, for which $\sim$ is the equivalence relation generated by

$$i_1 f(c) \sim i_2 g(c).$$

Example 2.2.8. If $f : C \rightarrow A$ and $g : C \rightarrow B$ are continuous function in **Top**, then their pushout is the same thing. The topology on $A \sqcup_C B$ is given by the quotient space of the disjoint union $A \sqcup B$, so that the canonical inclusions $i_1 : A \rightarrow A \sqcup_C B$, $i_2 : B \rightarrow A \sqcup_C B$ are continuous.

In particular, if $C = \text{point}$, then $A \sqcup_C B \cong A \vee B$ is the wedge product.

Exercise 2.2.9. Consider two diagrams in **Top**:

$$\begin{array}{ccc}
\mathbb{S}^n & \longrightarrow & \text{point} \\
\downarrow & & \\
\text{point} & &
\end{array}
\qquad
\begin{array}{ccc}
\mathbb{S}^n & \longrightarrow & \mathbb{D}^{n+1} \\
\downarrow & & \\
\text{point} & &
\end{array}$$

There is a natural weak equivalence between two diagrams. However, their pushouts are not weakly equivalent, since the pushout of the left one is a single point, while the right one is $\mathbb{S}^{n+1}$.

2.3 Limit and Colimit

2.3.1 Limit

Definition 2.3.1 (Limit). Let $F : \mathcal{J} \rightarrow \mathcal{C}$ be a functor. The **limit** of F is an object $L \in \mathcal{C}$ together with a collection of morphisms $\psi = \{\psi_X : L \rightarrow F(X) \mid X \in \mathcal{J}\}$ such

that for every $X, Y \in \mathcal{J}$ and $f \in \text{Mor}_{\mathcal{J}}(X, Y)$, the diagram

$$\begin{array}{ccc} & L & \\ \psi_X \swarrow & & \searrow \psi_Y \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

commutes. Moreover, the data (L, ψ) satisfies the following universal property: if $(L', \psi' = \{\psi'_X \mid X \in \mathcal{J}\})$ is another pair so that for every $X, Y \in \mathcal{J}$, and every $f \in \text{Mor}_{\mathcal{J}}(X, Y)$ the diagram

$$\begin{array}{ccc} & L' & \\ \psi'_X \swarrow & & \searrow \psi'_Y \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

commutes, then there exists a unique morphism $u : L' \rightarrow L$ in $\mathcal{C}$ such that for every $X, Y \in \mathcal{J}$, and $f \in \text{Mor}_{\mathcal{J}}(X, Y)$, the diagram

$$\begin{array}{ccc} & L' & \\ \psi'_X \swarrow & \downarrow u & \searrow \psi'_Y \\ & L & \\ \psi_X \swarrow & & \searrow \psi_Y \\ F(X) & \xrightarrow{F(f)} & F(Y). \end{array}$$

We write $\lim_{\mathcal{J}} F$ to denote the pair (L, ψ) .

Exercise 2.3.2. Show that $\lim_{\mathcal{J}} F$, if exists, is unique up to a unique isomorphism.

Example 2.3.3 (Product as Limit). One can realise product as limit. Suppose $\mathcal{J}$ is a category that

$$\text{Mor}_{\mathcal{J}}(X, Y) = \begin{cases} \{\text{id}_X\} & \text{if } X = Y, \\ \emptyset & \text{if else.} \end{cases}$$

Say $\text{Obj}(\mathcal{J}) = \{a, b\}$, and put $A := F(a), B := F(b) \in \mathcal{C}$. Then $\lim_{\mathcal{J}} F = A \times B$.

If $\mathcal{J}$ has more objects, then

$$\lim_{\mathcal{J}} F = \prod_{J \in \text{Obj}(\mathcal{J})} F(J).$$

Example 2.3.4 (Pullback as Limit). Let $\mathcal{J}$ be the category consists of three objects x, y, z ; and there are only two morphisms $f : x \rightarrow z$, $g : y \rightarrow z$ besides the identities.

Then $\lim_{\mathcal{J}} F$ the the pullback of the diagram

$$\begin{array}{ccc} & F(x) & \\ & \downarrow F(g) & \\ F(y) & \xrightarrow{F(f)} & F(z). \end{array}$$

2.3.1.1 Terminal Object

Let $\mathcal{J}$ be the category with one object $\emptyset$ and one morphism $\text{id}_{\emptyset}$. Then $\lim_{\mathcal{J}} F$ is an object T in $\mathcal{C}$, with the property that for every $X \in \mathcal{C}$, there is exactly one morphism $X \rightarrow T$. We call T a **terminal object** in $\mathcal{C}$.

Example 2.3.5. In **Set**, a singleton set is a terminal object.

Example 2.3.6. In **Vect**($\mathbb{F}$), the zero space is a terminal object.

Example 2.3.7. There are no terminal objects in **Field**. To see this, suppose $F \in \mathbf{Field}$ is a terminal object. Then from elementary field theory one knows the characteristic of F is either 0 or some prime number p .

If F has characteristic zero, then since F is a terminal object, there exists a field homomorphism $\varphi : \mathbb{Z}/p\mathbb{Z} \rightarrow F$ for every prime number p . In particular, $\varphi([1]) = 1_F$. Consequently,

$$\varphi([n]) = \underbrace{1_F + \cdots + 1_F}_n,$$

and

$$\underbrace{1_F + \cdots + 1_F}_p = 0,$$

contradicting the fact that F has characteristic zero.

If F has characteristic p , then there exists a field homomorphism $\mathbb{Q} \rightarrow F$. But this would imply F has characteristic zero by a similar argument.

Exercise 2.3.8. Show that a terminal object, if exists, is unique up to a unique isomorphism.

2.3.2 Colimit

Definition 2.3.9 (Colimit). Let $F : \mathcal{J} \rightarrow \mathcal{C}$ be a functor. The **colimit** of F is an object $M \in \mathcal{C}$ together with a collection of morphisms $\varphi = \{\varphi_X : F(X) \rightarrow M \mid X \in \mathcal{J}\}$ such that for every $X, Y \in \mathcal{J}$ and $f \in \text{Mor}_{\mathcal{J}}(X, Y)$, the diagram

$$\begin{array}{ccc} & M & \\ \varphi_X \nearrow & & \nwarrow \varphi_Y \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

commutes. Moreover, the data (M, φ) satisfies the following universal property: if $(M', \varphi' = \{\varphi'_X \mid X \in \mathcal{J}\})$ is another pair so that for every $X, Y \in \mathcal{J}$, and every $f \in \text{Mor}_{\mathcal{J}}(X, Y)$ the diagram

$$\begin{array}{ccc} & M' & \\ \varphi'_X \nearrow & & \nwarrow \varphi'_Y \\ F(X) & \xrightarrow{F(f)} & F(Y) \end{array}$$

commutes, then there exists a unique morphism $u : M \rightarrow M'$ in $\mathcal{C}$ such that for every $X, Y \in \mathcal{J}$, and $f \in \text{Mor}_{\mathcal{J}}(X, Y)$, the diagram

$$\begin{array}{ccc} & M' & \\ \varphi'_X \nearrow & \uparrow u & \nwarrow \varphi'_Y \\ & M & \\ \varphi_X \nearrow & & \nwarrow \varphi_Y \\ F(X) & \xrightarrow{F(f)} & F(Y). \end{array}$$

We write $\text{colim}_{\mathcal{J}} F$ to denote the pair (M, φ) .

Exercise 2.3.10. Show that $\text{colim}_{\mathcal{J}} F$, if exists, is unique up to a unique isomorphism.

Example 2.3.11 (Coproduct as Colimit). In the case as in Example 2.3.3, the colimit of the functor F is the coproduct $A \sqcup B$.

Example 2.3.12 (Pushout as Colimit). Let $\mathcal{J}$ be the category consists of three objects x, y, z ; and there are only two morphisms $f : x \rightarrow z$, $g : y \rightarrow z$ besides the identities.

Then $\text{colim}_{\mathcal{J}} F$ is the pushout of the diagram

$$\begin{array}{ccc}
 & F(x) & \xrightarrow{F(f)} F(y) \\
 & \downarrow F(g) & \\
 & F(z) & .
 \end{array}$$

2.3.2.1 Initial Object

Let $\mathcal{J}$ be the category with one object $\emptyset$ and one morphism $\text{id}_{\emptyset}$. Then $\text{colim}_{\mathcal{J}} F$ is an object I in $\mathcal{C}$, with the property that for every $X \in \mathcal{C}$, there is exactly one morphism $I \rightarrow X$. We call I an **initial object** in $\mathcal{C}$.

Example 2.3.13. In **Set**, the empty set is an initial object.

Example 2.3.14. In $\mathbf{Vect}(\mathbb{F})$, the zero space is an initial object.

Example 2.3.15 (Central Extension). Fix a group G , and an Abelian group A . A **central extension** of G by A is a pair (E, φ) , where E is a group containing A as a central subgroup, and $\varphi : E \rightarrow G$ is an epimorphism.

Central extensions of G by arbitrary Abelian groups form a category $\mathbf{CE}(G)$. A morphism from (E, φ) to (E', φ') is a commutative diagram

$$\begin{array}{ccccccccc}
 1 & \longrightarrow & A & \longrightarrow & E & \xrightarrow{\varphi} & G & \longrightarrow & 1 \\
 \parallel & & \parallel & & \downarrow \psi & & \parallel & & \parallel \\
 1 & \longrightarrow & A & \longrightarrow & E' & \xrightarrow{\varphi'} & G & \longrightarrow & 1.
 \end{array}$$

The initial object in $\mathbf{CE}(G)$ is called the **universal central extension** of G .

Exercise 2.3.16. Prove that there are no initial objects in **Field**.

Exercise 2.3.17. Show that an initial object, if exists, is unique up to a unique isomorphism.

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